Adaptive Clinical Recruitment: A Long-Horizon Benchmark for Sequential Trial Planning

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Abstract

Clinical recruitment creates delayed feedback by construction: screening, follow-up, site allocation, and budget decisions can affect outcomes weeks later, and those outcomes arrive under uncertainty. We present the **Adaptive Clinical Recruitment** benchmark, a deterministic 180-step environment for sequential trial-planning decisions with three public task difficulties, typed observations, delayed effects, milestone tracking, and site-level constraints. The core benchmark exposes 8 implemented action types and a 37-dimensional numeric feature vector used by the trainable baselines. The repository also contains four repo baselines aimed at different long-horizon behaviors: hierarchical subgoals with hindsight relabeling, milestone-based shaping, multi-scale temporal aggregation, and episodic memory. We regenerate all reported results from the corrected evaluation path and run a fresh 5-seed sweep (30 training episodes per seed, 15 evaluation episodes per seed). The highest mean score is 0.2215, but no pairwise comparison reaches $p < 0.05$; under the current training budget, the baseline suite is best read as a check on environment and evaluation behavior rather than as evidence that one approach clearly wins. The submission artifact includes code, regenerated charts, and a passing integration suite for all three tasks.

1 Introduction

Patient recruitment is one of the main reasons clinical trials miss timelines and overspend budget (Fogel, 2018). Unlike short-horizon benchmark domains, recruitment decisions are shaped by delayed conversions, dropout risk, site capacity, and non-stationary candidate pools. A benchmark in this space should therefore test more than immediate reward optimization: it should expose planning state, require action-specific candidate selection, and make it easy to reproduce both environment behavior and evaluation outputs.

This paper presents an anonymized NeurIPS E&D submission package for the **Adaptive Clinical Recruitment** repository. The aim is not to argue for a new best-performing algorithm. The contribution is a benchmark package whose claims stay close to the code, regenerated artifacts, and validation outputs currently present in the repository.

Our main contributions are:

1. A deterministic 180-step recruitment environment with three task levels, delayed effects, milestone tracking, site-level metrics, token accounting, and typed observations.
2. A clearly scoped action interface with 8 implemented action types and action-specific candidate pools for screening, recontact, and allocation.
3. Four repo baselines that instantiate distinct long-horizon mechanisms using a shared pure-NumPy actor-critic backbone.

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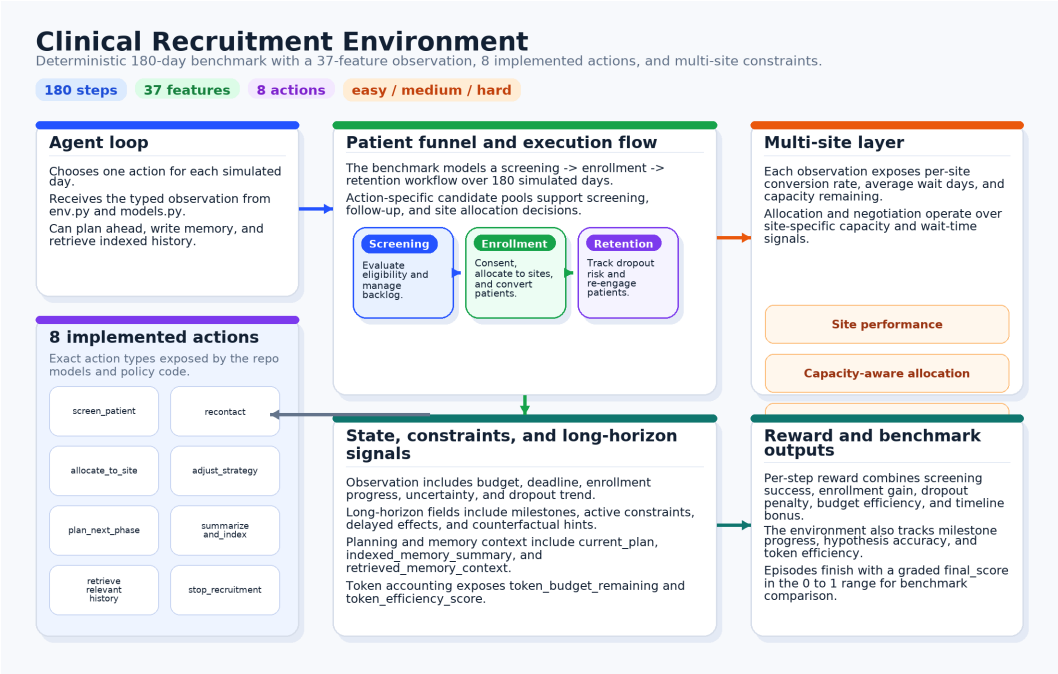

Figure 1: Programmatic overview of the current benchmark implementation. The figure is regenerated from repository-backed labels and reflects the current 8-action interface.

Table 1: Task families used by the repository and the sweep runner.

Task	Sites	Budget	Target	Main pressure
`easy_bench`	1	\$120K	80	Basic funnel dynamics
`medium_bench`	3	\$150K	120	Multi-site coordination
`hard_bench`	5	\$100K	150	Budget and retention pressure

4. A fresh 5-seed sweep, regenerated charts, and 30/30 passing integration checks across the task suite.

2 Benchmark Design

2.1 Environment Overview

The environment simulates a 180-day recruitment window in which an agent manages a patient funnel from screening through enrollment and retention. Figure 1 summarizes the core benchmark surfaces. At each step, the environment returns a typed observation containing both structured state and a numeric feature representation used by the trainable baselines.

The benchmark provides three public tasks with increasing difficulty (Table 1). The harder tasks tighten budget, increase target enrollment, and add more site coordination pressure.

2.2 Observations, Actions, and Rewards

The training stack extracts a 37-dimensional numeric feature vector from each observation. That vector is built from timestamp, budget, deadline, enrollment progress, funnel counts, milestone status, constraint flags, uncertainty components, memory summaries, and counterfactual rollout hints. The full typed observation also carries structured fields such as `available_patients`, `recontact_candidates`, `allocation_candidates`, `site_performance`, `current_plan`, and `retrieved_memory_context`.

Table 2: Implemented action interface. Candidate selection is action-specific.

Action	Primary role
`screen_patient`	Screen a candidate from `available_patients`.
`recontact`	Re-engage a candidate from `recontact_candidates`.
`allocate_to_site`	Allocate a consented candidate from `allocation_candidates`.
`adjust_strategy`	Change outreach, criteria, or site focus.
`plan_next_phase`	Update high-level plan state for the next phase.
`summarize_and_index`	Write a compact entry into indexed memory.
`retrieve_relevant_history`	Query indexed memory for current bottlenecks.
`stop_recruitment`	Terminate the episode early.

Table 3: Repo baselines and the long-horizon mechanism each emphasizes.

Baseline	Main mechanism
HCAPO	Repo baseline with planner/executor separation and hindsight relabeling, borrowing the relabeling idea from HER (Andrychowicz et al., 2017).
MiRA	Repo baseline with a learned milestone potential term, borrowing the reward-shaping idea from policy-invariant shaping (Ng et al., 1999).
KLong	Repo baseline with overlapping temporal windows and TD(λ)-style credit flow (Sutton and Barto, 2018).
MemexRL	Repo baseline with gated episodic writes and similarity-based retrieval, borrowing from neural episodic control (Pritzel et al., 2017).

The current action interface contains 8 implemented action types (Table 2). This is the interface used by `models.py`, `env.py`, and `training/neural_policy.py`; earlier drafts that listed additional actions were incorrect and are not used in this submission.

Rewards combine screening success, enrollment gain, dropout penalty, budget efficiency, timeline pressure, milestone events, hypothesis accuracy, and token efficiency. Delayed effects are queued explicitly, so actions can trigger downstream consequences several steps later.

2.3 Core Scope Versus Experimental Extras

The repository contains additional research helpers beyond the core benchmark path, including offline analysis modules, counterfactual simulation utilities, curriculum managers, and reporting stubs. In this paper, we treat only the following as core benchmark components: `env.py`, `models.py`, the four trainable baselines, `experiments/full_sweep.py`, `experiments/reproducibility.py`, and the test scripts used for validation. We do *not* treat modules such as `training/async_rl.py`, `training/progressive_rl.py`, or `experiments/appendix_report.py` as validated benchmark contributions.

3 Repository Baselines and Evaluation Protocol

3.1 Repo Baselines

The repository includes four trainable baselines built on a shared pure-NumPy actor-critic implementation. We describe them as repo baselines rather than strict reproductions of external named methods.

All four baselines use the same 37-dimensional feature extractor from `training/neural_policy.py`. Before regenerating the reported results, we corrected the evaluation path so that `recontact` and `allocate_to_site` choose patients from their action-specific candidate pools instead of from the unscreened screening pool. All results reported below were regenerated after that correction.

Table 4: Fresh 5-seed sweep regenerated from the corrected evaluation path. Scores are final environment grades in $(0, 1)$.

Baseline	Mean	Std	95% CI
HCAPO	0.2215	0.0127	[0.2100, 0.2303]
KLong	0.2152	0.0222	[0.1977, 0.2286]
MemexRL	0.2148	0.0270	[0.1943, 0.2352]
MiRA	0.2094	0.0095	[0.2023, 0.2165]

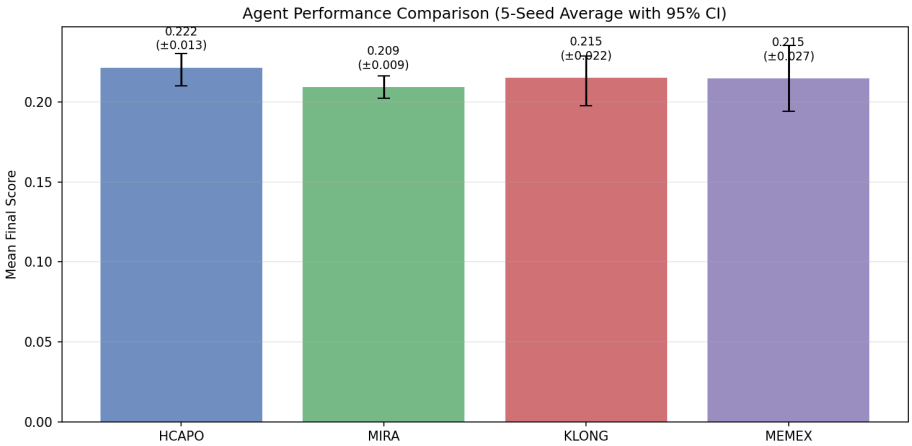

Figure 2: Regenerated agent comparison chart copied directly from the fresh sweep output.

3.2 Training and Compute

The fresh sweep uses 5 seeds: {1, 7, 21, 42, 123}. For each seed and baseline, `experiments/full_sweep.py` runs 30 training episodes while cycling across the three tasks, then evaluates on 5 episodes per task (15 evaluation episodes total). Pairwise comparisons use paired t-tests, Wilcoxon signed-rank tests, Cohen's d, and bootstrap 95% confidence intervals computed in `experiments/reproducibility.py`.

All reported experiments were executed locally on a 12th Gen Intel Core i5-1245U CPU with 16 GB RAM, Python 3.14, and no GPU. The full 4-baseline $\times$ 5-seed sweep completed in approximately 295.5 seconds (4.9 minutes) end-to-end.

4 Results

Table 4 and Figure 2 summarize the regenerated 5-seed sweep. HCAPO attains the highest mean score, but the four baselines remain tightly clustered and their confidence intervals overlap.

No pairwise comparison reaches $p < 0.05$. The smallest p-value is 0.1823 for HCAPO versus MiRA. HCAPO versus KLong, a comparison overstated in older drafts, now yields $p = 0.3849$. The conservative reading is that all four baselines make partial progress on the benchmark, but the repository does not yet provide statistical evidence that one baseline is clearly ahead of the others.

The integration suite passes 10/10 checks on each of `easy_bench`, `medium_bench`, and `hard_bench`, for a total of 30/30 checks. These checks validate observation structure, site metrics, milestone exposure, plan and memory fields, token-budget state, multi-step execution, action diversity, and non-zero reward accumulation.

5 Interpretation and Limitations

The current repository is strongest as an evaluation package: the environment interface is typed and deterministic, the action construction path is now consistent with the observation schema, and the reported figures can be regenerated from a short CPU run. The benchmark is less mature as an empirical leaderboard. Baseline means are close, variance remains non-trivial, and the present training budget is small.

Several limitations matter for reviewers. First, the environment is synthetic and should not be treated as a deployment proxy for actual clinical operations. Second, some repository modules remain scaffolds or offline research helpers rather than validated benchmark components. Third, the training stack uses simple pure-NumPy neural networks rather than stronger modern RL backbones. Finally, because the baselines are clustered, stronger evaluation protocols (more seeds, larger training budgets, and ablations over individual state channels) are still needed.

The broader impact is mixed. A reproducible workflow benchmark can help stress-test long-horizon agents for planning, memory, and delayed effects, but any real-world use in healthcare operations would raise questions about fairness, patient autonomy, and inappropriate automation. For that reason, we position this artifact as a research benchmark only.

6 Conclusion

The **Adaptive Clinical Recruitment** repository provides a reproducible long-horizon benchmark package with a deterministic 180-step environment, a corrected 8-action interface, four repo baselines, fresh 5-seed sweep outputs, and passing integration checks across three task difficulties. The current evidence does *not* show statistically significant separation among the baselines, which is itself informative: future work on this benchmark should emphasize stronger training protocols, sharper ablations, and a continued separation between core benchmark claims and experimental scaffolds.

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Table 5: Complete pairwise comparisons from the fresh 5-seed sweep. None reach $p < 0.05$.

Comparison	Mean diff	p-value	Cohen's d
HCAPO vs MiRA	0.0121	0.1823	1.076
HCAPO vs KLong	0.0063	0.3849	0.348
HCAPO vs MemexRL	0.0067	0.6370	0.319
MiRA vs KLong	-0.0058	0.6674	-0.340
MiRA vs MemexRL	-0.0053	0.6656	-0.264
KLong vs MemexRL	0.0004	0.9756	0.018

A Pairwise Statistical Tests

B Integration Checks and Runtime

Table 6: Integration checks regenerated from `data/sweep_results/integration_tests.json`.

Task	Checks passed	Steps run	Total reward
`easy_bench`	10/10	20	4.6836
`medium_bench`	10/10	20	2.6390
`hard_bench`	10/10	20	2.1074

The repository validation scripts currently report 76 passing checks in `test_env.py`, 43 in `test_agents.py`, 109 in `test_research_modules.py`, and 77 in `test_local_serving.py`. The paper claims rely on the core benchmark and fresh sweep outputs, not on the existence of every auxiliary research helper.

NeurIPS Paper Checklist

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   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction limit claims to the implemented 8-action benchmark, the fresh 5-seed sweep, and the reproducibility package described in Sections 1–4.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

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   Justification: Section 5 discusses synthetic dynamics, small training budgets, clustered baseline performance, and the distinction between core benchmark code and auxiliary scaffolds.

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   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

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   Justification: Sections 3–4 specify seeds, episode counts, evaluation structure, compute setup, and the exact repository scripts used to regenerate the sweep.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The benchmark, sweep scripts, regenerated figures, and result artifacts are present in the repository; the paper cites the relevant file paths and appendix tables.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

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   Justification: Section 3 gives the seed set, episode counts, evaluation count per task, shared pure-NumPy actor-critic backbone, and the sweep runner used for all reported numbers.

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   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

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   Justification: Section 4 reports standard deviations, bootstrap 95% confidence intervals, and full pairwise statistical tests; Appendix A lists all pairwise comparisons.

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Answer: [Yes]

Justification: The paper cites the external benchmark and RL references used for context, and the repository itself is distributed under an MIT license noted in the project documentation.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: The new asset is the benchmark package itself; the repository includes environment code, sweep outputs, charts, and documentation describing the current implemented scope.

14. **Crowdsourcing and research with human subjects**

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

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Justification: The benchmark is synthetic and no human subject study was conducted.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?

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